SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

DuoLogic

Project / Solution Name

Noxera

Name (Leader)

Chang Zhi Qing

Email & Phone (Leader)

changzhiqing1996@gmail.com & 0192723941

University Affiliation(s)

Universiti Tunku Abdul Raman (Sungai Long)

Live Prototype URL

https://nexora-2.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/Zhi-Qing-6272/SDC_Hackathon_Submission

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Nexora uses a React/TypeScript/Vite browser interface with Clerk sign-in. The user creates a skills,experience and interests profile; the Express REST API validates requests and stores account-owned records in PostgreSQL through Drizzle ORM. Browser-side code extracts text from TXT/MD/DOCX CVs (not PDFs), computes explainable skill overlap, matched and missing skills, role scores, scenario rankings and learning steps. Raw CV text is not sent to the server.

The Express server retrieves public remote-job listings from the Himalayas API. A 6-hour in-memory cache feeds market-signal counts and live vacancy matching; UI panels display a source and fetched date. Curated course and US BLS links provide external guidance, not live course API calls. Organisation features include team skill-gap matrices and synthetic sample employees.

The Help chat uses built-in FAQs by default, or calls Gemini/Claude when configured with server-side credentials. It is separate from the deterministic career matcher. Clerk manages account credentials; optional SMTP sends invites.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.1.4 / TS ~5.9	Browser UI and typed logic	MIT / Apache 2.0
Vite ^7.3.2 / Tailwind ^4.1.14	Build, CSS styling	MIT
Express ^5.2.1 / Node 24	Server APIs	MIT
Drizzle ORM ^0.45.2; pg	PostgreSQL data access	Apache-2.0 / MIT
Clerk React / Express SDK	Auth, sessions, accounts	MIT SDK + service terms
Radix UI / Lucide / Recharts	Accessible UI, icons, charts	MIT / ISC / MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Himalayas jobs API	Remote vacancies; dated skill counts	free
Clerk identity service	Sign-in, account verification	free
Google Gemini API	Help chat only	free
Anthropic Messages API	Help chat only	free
SMTP email provider	Team invitation email if configured	free
External courses	Outbound links; not API calls	free
US BLS / O*NET	Cited reference links, no API	free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit / Node runtime	Website and backend deployment,	Core
PostgreSQL	Store user accounts, skills and team data	Free
Clerk hosted identity	User authentication and sessions	Free
Himalayas API	Provide remote job-market data	Free public API
Gemini / Google	Optional AI Help assistant	Free
Claude / Anthropic	Optional AI Help assistant	Free
SMTP / provider TBD	Optional invitation emails	Free

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini	Google	pre-trained	Google AI terms
Claude	Anthropic	pre-trained	Anthropic terms

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Himalayas job ads	himalayas.app	Public live jobs	API ToS	N
US BLS projections	bls.gov	Public reference	BLS attribution	N
Learning links	Provider websites	Curated links	Each provider	N
O*NET search links	onetonline.org	Reference links	Website terms	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit agent	Website scaffolding/code assistance	major
ChatGPT	Development advice; forms/docs	moderate
Claude	README: admin translations	major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team must review AI-assisted code and translations against the actual source and run feature tests with synthetic data. The repository documents deterministic career-match formulas, typed API validation and authenticated database routes. A source review also found unsupported PDF CV parsing, limited global-remote labour signals and optional, not guaranteed, AI Help connectivity.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	$20 credits	assist in performing security check before publishing the website

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was used to develop, test, and deploy Nexora. It helped identify potential security vulnerabilities, troubleshoot errors, and publish the website. However, Replit is not essential to Nexora's core functionality, as the application could also be developed using other tools, such as VS Code, and deployed through alternative cloud platforms. Therefore, Replit was beneficial for simplifying development and deployment but could be replaced without significantly affecting the system's core features.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The source stores signed-in users' names, roles, experience notes, skills, interests, goals and
learning completion in account-owned PostgreSQL records; Clerk manages sign-in credentials. Companies may store registration/contact data and consenting employees' shared work skills. CV text is processed locally and is not saved or uploaded; only selected profile fields are saved. Team invites may use optional SMTP. Help chat sends at most the last 12 typed messages plus page, account type and language to Gemini/Claude only when configured; its messages are not saved by Nexora, but provider retention depends on provider terms. Protected endpoints require Clerk sessions and server-side ownership checks.
Workspace deletion is implemented; no fixed server retention period is asserted.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Himalayas shows a limited worldwide remote-jobs sample, not representative Malaysian demand.
The 6-hour cache may become stale; upstream API outages, schema changes and rate limits can interrupt updates. Skills are inferred from titles/categories/ excerpts, producing false matches or missed skills. Rule-based role scores and moderate/fast disruption scenarios are planning heuristics, not validated forecasts; US BLS figures do not describe Malaysia. Listed courses and pricing may change. CV parsing supports TXT/MD/DOCX but not PDF in this build. Optional Gemini/Claude Help may hallucinate and expose typed chat messages to its provider if enabled.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

Zhi

[Date]

22/10/2026